

Science question paper

SUBJECT

Science

CLASS

10

TOPIC

Electricity

MAXIMUM MARKS

50

TIME

1 Hour

General Instructions:

1. The question paper consists of five sections A, B, C, D and E.
2. All questions are compulsory.
3. Section A consists of multiple choice questions.
4. Section E consists of case-study based questions.
5. Diagrams should be neat and properly labelled wherever required.

Section A - Multiple Choice Questions

Choose the correct option.

10 × 1 = 10

1. What is the phenomenon called when an electric current passes through a conductor and produces heat? [1]
(A) Magnetic effect (B) Chemical effect
(C) Heating effect (D) Luminous effect
2. Three resistors of 2 Ω, 3 Ω, and 6 Ω are connected in parallel. What is their equivalent resistance? [1]
(A) 11 Ohm (B) 6 Ohm
(C) 1 Ohm (D) 0.5 Ohm
3. Which of the following describes electric power? [1]
(A) Total energy (B) Rate of energy
(C) Energy per charge (D) Force on charge
4. If a conductor's length is doubled and its area of cross-section is halved, how does its resistance change? [1]
(A) Halved (B) Doubled
(C) Four times (D) Unchanged
5. What is the SI unit of resistivity? [1]
(A) Ohm (B) Ohm-meter
(C) Siemens (D) Volt/Ampere
6. When resistors are connected in parallel, what remains the same across them? [1]
(A) Current (B) Resistance
(C) Power (D) Potential difference
7. Why is an electric fuse typically made of a material with a low melting point? [1]

- | | |
|-------------------------|----------------------|
| (A) Conduct electricity | (B) Melt easily |
| (C) Be strong | (D) Resist corrosion |

8. If the potential difference across a resistor is doubled, what happens to the current flowing through it, assuming resistance is constant? [1]

- | | |
|------------------------|-----------------------|
| (A) Current halves | (B) Current doubles |
| (C) Current quadruples | (D) Current unchanged |

9. What is the SI unit of electric current? [1]

- | | |
|------------|-------------|
| (A) Volt | (B) Ohm |
| (C) Ampere | (D) Coulomb |

10. Which principle is utilized in the working of an electric fuse? [1]

- | | |
|---------------------|---------------------|
| (A) Magnetic effect | (B) Chemical effect |
| (C) Heating effect | (D) Luminous effect |

Section B - Very Short Answer Questions

Answer the following questions briefly.

5 × 2 = 10

11. Differentiate between series and parallel combinations of resistors based on how the total resistance changes when more resistors are added. [2]

12. Why is it generally not advisable to connect household appliances in series? [2]

13. State two advantages of connecting electrical appliances in parallel in a domestic circuit. [2]

14. Define electric potential difference and state its SI unit. [2]

15. Consider two identical electrical heaters. If one is connected in series and the other in parallel to the same power supply, which one would draw more current and why? [2]

Section B - Short Answer

Answer in 2-3 sentences.

5 × 3 = 15

16. Derive the expression for the equivalent resistance when three resistors R1, R2, and R3 are connected in series. [3]

17. Describe the working principle of an electric fuse. [3]

18. Discuss the advantages of parallel circuits over series circuits in domestic wiring. [3]

19. Differentiate between series and parallel combinations of resistors based on current flow and potential difference. [3]

20. Derive the expression for the equivalent resistance when two resistors R1 and R2 are connected in series, using Ohm's Law. [3]

Section D - Long Answer Questions

Answer the following questions in detail.

$3 \times 5 = 15$

21. An electric heater is rated 1500 W, 220 V. Calculate the current drawn by the heater and the power consumed if it operates at 200 V. [5]

22. State Joule's Law of heating. On what factors does the heat produced in a resistor depend? [5]

23. Describe the heating effect of electric current and state Joule's Law of heating. Explain two practical applications of the heating effect of electric current. [5]

Answer Key

- Q1.** Heating effect
- Q2.** 1 Ohm
- Q3.** Rate of energy
- Q4.** Four times
- Q5.** Ohm-meter
- Q6.** Potential difference
- Q7.** Melt easily
- Q8.** Current doubles
- Q9.** Ampere
- Q10.** Heating effect
- Q11.** In a series combination, adding more resistors increases the total resistance. In a parallel combination, adding more resistors decreases the total resistance.
- Q12.** If household appliances are connected in series, they would not receive the full supply voltage, and if one appliance fails, the entire circuit would break, stopping all other appliances from working.
- Q13.** Two advantages are: (1) Each appliance gets the full supply voltage. (2) If one appliance stops working, others continue to operate normally.
- Q14.** Electric potential difference between two points in an electric circuit is defined as the work done to move a unit charge from one point to the other. Its SI unit is Volt (V).
- Q15.** The heater connected in parallel would draw more current. This is because in a parallel circuit, the resistance is lower than in a series circuit (assuming the other heater is also in the circuit for comparison), and thus, by Ohm's Law ($I=V/R$), a lower resistance leads to higher current for the same voltage.
- Q16.** When resistors are connected in series, the current flowing through each resistor is the same, but the potential difference across each resistor is different. According to Ohm's Law, $V_1 = IR_1$, $V_2 = IR_2$, $V_3 = IR_3$. The total potential difference V across the combination is the sum of the potential differences across individual resistors: $V = V_1 + V_2 + V_3$. Substituting the values, $V = IR_1 + IR_2 + IR_3 = I(R_1 + R_2 + R_3)$. If R_s is the equivalent resistance, then $V = I R_s$. Therefore, $I R_s = I(R_1 + R_2 + R_3)$, which simplifies to $R_s = R_1 + R_2 + R_3$.
- Q17.** An electric fuse is a safety device that protects electrical circuits and appliances from damage due to excessive current. It works on the heating effect of electric current. It consists of a short piece of wire made of an alloy with a low melting point, such as tin-lead alloy. When the current in the circuit exceeds a safe limit, the fuse wire heats up due to the Joule heating effect ($H=I^2Rt$) and melts, breaking the circuit and stopping the flow of current, thus protecting the appliances.
- Q18.** In parallel circuits, if one appliance stops working, others continue to operate normally, unlike series circuits where the entire circuit breaks. Appliances in parallel circuits receive the full supply voltage (e.g., 220V), ensuring they operate at their rated power, whereas in series, voltage divides. Also,

parallel circuits allow different appliances with different current requirements to be connected without affecting each other's operation.

Q19. In a series combination, the current is the same through all resistors, but the potential difference across each resistor is different and adds up to the total potential difference. In contrast, in a parallel combination, the potential difference across each resistor is the same as the total potential difference, while the total current divides among the resistors.

Q20. When two resistors R_1 and R_2 are connected in series, the same current (I) flows through both. The total potential difference (V) across the combination is the sum of the potential differences across R_1 (V_1) and R_2 (V_2). According to Ohm's Law, $V_1 = IR_1$ and $V_2 = IR_2$. So, $V = V_1 + V_2 = IR_1 + IR_2 = I(R_1 + R_2)$. If R_s is the equivalent resistance of the series combination, then $V = I R_s$. Equating the two expressions for V , we get $I R_s = I(R_1 + R_2)$, which simplifies to $R_s = R_1 + R_2$.

Q21. Given: Power (P) = 1500 W, Voltage (V) = 220 V

Step 1: Calculate the current drawn at 220 V.

$$P = V \cdot I$$

$$I = P / V = 1500 \text{ W} / 220 \text{ V} = 6.82 \text{ A (approx)}$$

Step 2: Calculate the resistance of the heater.

$$P = V^2 / R$$

$$R = V^2 / P = (220 \text{ V})^2 / 1500 \text{ W} = 48400 / 1500 = 32.27 \text{ ohms (approx)}$$

Step 3: Calculate the current drawn if it operates at 200 V.

$$I' = V' / R = 200 \text{ V} / 32.27 \text{ ohms} = 6.20 \text{ A (approx)}$$

Step 4: Calculate the power consumed if it operates at 200 V.

$$P' = V' I' = 200 \text{ V} \cdot 6.20 \text{ A} = 1240 \text{ W (approx)}$$

OR

$$P' = (V')^2 / R = (200 \text{ V})^2 / 32.27 \text{ ohms} = 40000 / 32.27 = 1239.85 \text{ W (approx)}$$

Final Answer: The current drawn by the heater at 220 V is approximately 6.82 A. If it operates at 200 V, the current drawn is approximately 6.20 A and the power consumed is approximately 1240 W.

Q22. Consider three resistors R_1 , R_2 , and R_3 connected in series across a voltage source V .

Step 1: Understand the characteristics of a series circuit.

In a series circuit, the current (I) flowing through each resistor is the same, but the potential difference (voltage) across each resistor is different.

Step 2: Apply Ohm's Law to each resistor.

According to Ohm's Law, $V = I \cdot R$.

Potential difference across R_1 : $V_1 = I \cdot R_1$

Potential difference across R_2 : $V_2 = I \cdot R_2$

Potential difference across R_3 : $V_3 = I \cdot R_3$

Step 3: Relate the total potential difference to individual potential differences.

The total potential difference (V) across the combination is the sum of the potential differences across individual resistors:

$$V = V_1 + V_2 + V_3$$

Step 4: Substitute the expressions from Ohm's Law.

$$V = (I R_1) + (I R_2) + (I \cdot R_3)$$

$$V = I \cdot (R_1 + R_2 + R_3)$$

Step 5: Define equivalent resistance.

If R_s is the equivalent resistance of the series combination, then according to Ohm's Law for the entire circuit:

$$V = I \cdot R_s$$

Step 6: Equate the expressions for V .

$$I R_s = I (R_1 + R_2 + R_3)$$

Step 7: Derive the final expression.

Dividing both sides by I (since I is not zero):

$$R_s = R_1 + R_2 + R_3$$

Thus, the equivalent resistance of resistors connected in series is the sum of their individual resistances.

Q23. The heating effect of electric current is the phenomenon where heat is produced in a conductor when electric current flows through it. This occurs because the moving electrons in the current collide with the atoms and ions of the conductor, transferring some of their kinetic energy to these particles. This increased vibrational energy of the conductor's particles manifests as heat.

Joule's Law of Heating states that the heat (H) produced in a resistor is directly proportional to the square of the current (I) flowing through it, the resistance (R) of the conductor, and the time (t) for which the current flows. Mathematically, it is expressed as:

$$H = I^2 R t$$

Practical applications of the heating effect of electric current:

Electric Heater/Geyser: These appliances use a high resistance wire (like nichrome) which heats up significantly when current passes through it, converting electrical energy into heat energy to warm water or air.

Electric Fuse: A fuse wire has a low melting point and high resistance. When a current exceeding a safe limit flows through the circuit, the fuse wire heats up rapidly and melts, breaking the circuit and protecting appliances from damage.